

Sea Ice Classification

Deep Fusion Workflow

Project Overview and Implementation Plan

This document describes the end-to-end workflow for classifying Antarctic sea ice into three surface classes (thick ice, thin ice, and open water) by fusing Sentinel-2 multispectral optical imagery with ICESat-2 ATL03 photon-altimetry data through a multimodal deep neural network. The architecture combines a U-Net image encoder (ResNet-18 backbone) with a recurrent LSTM photon encoder, merging both modalities through a deep feature-level fusion stage with Squeeze-and-Excitation channel recalibration. The fused model achieves a mean Intersection-over-Union (mIoU) of 0.9010 and a macro-averaged F1 score of 0.9468 on the geographically held-out test tile T03CWT.

Key Result: Deep Fusion mIoU = 0.9010 | Macro F1 = 0.9468 | Pixel Accuracy = 95.3%

1. Data Overview and Objectives

The sea ice classification task is formulated as a per-pixel prediction problem. Each pixel is classified into one of three surface types using paired optical imagery and along-track photon-altimetry measurements. Training is performed on tiles T02CNA and T02CNC; evaluation is done on the geographically separated tile T03CWT to test geographic generalization.

Sentinel-2 Level-1C: Optical RGB imagery at 10 m resolution over the Ross Sea region. Patches of 128 x 128 pixels are extracted centered on each labeled ICESat-2 photon point. Patches capture texture, reflectance, and spatial context relevant to surface-type identification.

ICESat-2 ATL03: Geolocated photon point clouds aggregated to 10-meter along-track segments. For each labeled location a sliding window of five consecutive segments is assembled. Eight engineered features are extracted per segment: mean and median corrected height, height asymmetry, relative elevation, height standard deviation, photon-count height, photon count, and background rate.

Ground-truth masks: Per-pixel labels generated by an HSV color-thresholding pipeline with cloud and shadow removal. Class encoding: red = thick ice, blue = thin ice, green = open water.

Source	Description	Resolution
Sentinel-2 L1C	Optical RGB imagery, Ross Sea	10 m
ICESat-2 ATL03	Photon altimetry, along-track segments	10 m along-track
Ground-truth masks	HSV-thresholded per-pixel labels	10 m

2. Baseline Models

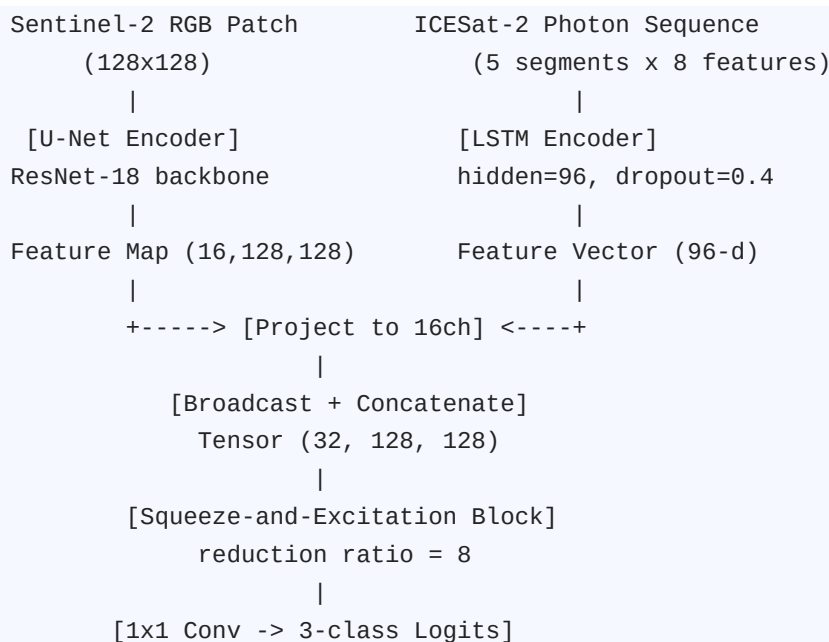
Two unimodal baselines establish the contribution of each data source independently before fusion.

- U-Net baseline:** A U-Net with ResNet-18 encoder pretrained on ImageNet processes each 128 x 128 optical patch. The decoder restores spatial resolution to produce pixel-wise class predictions. Test mIoU on T03CWT: 0.8704.
- LSTM baseline:** A single-layer LSTM (hidden dimension 96, dropout 0.4) processes the five-segment photon sequence and produces a per-point class prediction via a softmax head. Training uses categorical focal loss (alpha=[0.05, 0.45, 0.60], gamma=2.0) selected by a 21-run hyperparameter sweep. Test mIoU on T03CWT: 0.6978.

3. Deep Fusion Architecture

The deep fusion model combines both encoders at the feature level rather than combining their output predictions (late fusion).

- Image branch:** U-Net with ResNet-18 encoder. The decoder produces a 16-channel spatial feature map of shape (16, 128, 128) capturing multi-scale optical texture.
- Photon branch:** Single-layer LSTM over a 5-segment window. The final hidden state is a 96-dimensional vector summarizing photon-height context along the satellite track.
- Fusion stage:** The 96-d photon vector is projected to 16 channels and broadcast across the spatial grid (16, 128, 128), then concatenated with the U-Net feature map to form a 32-channel tensor. A Squeeze-and-Excitation block (reduction ratio 8) performs channel-wise recalibration. A final 1 x 1 convolution outputs three-class per-pixel logits.
- Fine-tuning:** The pretrained photon branch is fine-tuned at 0.1x the base learning rate during fusion training, allowing adaptation while preserving learned photon representations.



|
[Softmax: thick ice / thin ice / water]

4. Implementation Workflow

Step 1: Data Preprocessing

Download Sentinel-2 Level-1C GeoTIFF scenes and ICESat-2 ATL03 photon CSV files. Normalize Sentinel-2 pixel values per channel. Remove invalid or cloud-contaminated pixels. Standardize photon-altimetry features (z-score). Generate per-pixel ground-truth masks using HSV color thresholding with cloud and shadow removal.

Step 2: Patch Extraction and Sequence Assembly

For each labeled ICESat-2 photon point, extract a 128 x 128 optical patch centered at (pix_x, pix_y). Assemble LSTM input sequences by gathering the five nearest consecutive along-track segments. Extract the eight engineered features per segment. Pair each sequence with its corresponding optical patch.

Step 3: Train Unimodal Baselines

Train the U-Net optical baseline and the LSTM photon baseline independently. Use focal loss with per-class alpha weights to prevent collapse onto dominant thick-ice class. Run 21-configuration hyperparameter sweep for the LSTM branch to select alpha, gamma, hidden size, learning rate, dropout, and sequence length.

Step 4: Train Deep Fusion Model

Initialize the fusion model with the pretrained photon branch. Set photon-branch learning rate to 0.1x of the base rate. Concatenate U-Net and LSTM features via the Squeeze-and-Excitation fusion module. Train end-to-end for 30 epochs on tiles T02CNA and T02CNC.

Step 5: Evaluate on Held-Out Tile

Evaluate all models on geographically separated tile T03CWT. Report mIoU, pixel accuracy, macro F1, and per-class precision/recall/F1. Compute row-normalized confusion matrices. Compare against unimodal baselines.

Step 6: Ablation Study

Compare four fusion variants: (v2) random-init photon branch (collapsed), (v3) larger random-init branch (hidden=128), (v4) pretrained frozen branch, (deep_fusion) pretrained fine-tuned branch. Confirm that transfer learning outperforms increased model capacity.

5. Results

Model	Input	mIoU	Pixel Acc.	Macro F1
U-Net	Sentinel-2 optical	0.8704	0.9429	N/A
LSTM	ICESat-2 photon	0.6978	0.9594	0.8080
Deep Fusion	optical + photon	0.9010	0.9530	0.9468

6. Conclusion

The deep fusion of Sentinel-2 optical imagery and ICESat-2 ATL03 photon-altimetry data achieves robust per-pixel Antarctic sea ice classification. The U-Net encoder provides rich multi-scale spatial features while the LSTM summarizes photon-height context along the satellite track. The Squeeze-and-Excitation fusion stage learns to dynamically weight both modalities, yielding mIoU 0.9010 and macro F1 0.9468 on the held-out tile -- improving thin-ice IoU by +4.5 pp and open-water IoU by +3.6 pp over the optical-only baseline.

The key design insight is that fine-tuning a pretrained photon branch at a reduced learning rate outperforms training a larger branch from scratch, demonstrating that transferred photon representations are more valuable than additional capacity.

Future work may incorporate additional spectral bands from Sentinel-2, attention-based fusion mechanisms, or extend coverage to other polar regions.